In [ ]:
```
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# Roll_no :09
```

In [1]:
```
import numpy as np
import pandas as pd

df=pd.read_csv('data_large.csv')
df.head()
```

Out[1]:

	Gender	Age	Income
0	Female	45	73608
1	Female	33	52160
2	Male	33	48597
3	Female	26	74835
4	Male	40	75057

In [2]:
```
df.describe()
```

Out[2]:

	Age	Income
count	80.000000	80.000000
mean	38.737500	59197.700000
std	8.789729	18409.083296
min	24.000000	30025.000000
25%	31.750000	42442.500000
50%	39.000000	59300.500000
75%	46.000000	77052.500000
max	53.000000	86949.000000

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 80 entries, 0 to 79
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   Gender  80 non-null     object
 1   Age     80 non-null     int64 
 2   Income  80 non-null     int64 
dtypes: int64(2), object(1)
memory usage: 2.0+ KB
```

In [10]:
```
summary = df.groupby('Gender')['Income'].agg(
    mean='mean',
    median='median',
    minimum='min',
    maximum='max',
```

```
    std_dev='std',
    variance='var'
)
```

In [12]:
```
summary.round(2)
```

Out[12]:

Gender	mean	median	minimum	maximum	std_dev	variance
Female	61526.7	62094.5	30267	86949	18220.21	3.319761e+08
Male	56868.7	50235.0	30025	86568	18530.40	3.433756e+08

In [13]:
```
dff=pd.read_csv('Iris.csv')
dff
```

Out[13]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [14]:
```
dff.head()
```

Out[14]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [16]:
```
dff.dtypes
```

Out[16]:
```
Id                 int64
SepalLengthCm    float64
SepalWidthCm     float64
PetalLengthCm    float64
PetalWidthCm     float64
Species           object
dtype: object
```

In [17]:
```
dff.describe()
```

Out[17]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

In [19]:
```
dff.select_dtypes('number')     # displaying numerical values
```

Out[19]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	1	5.1	3.5	1.4	0.2
1	2	4.9	3.0	1.4	0.2
2	3	4.7	3.2	1.3	0.2
3	4	4.6	3.1	1.5	0.2
4	5	5.0	3.6	1.4	0.2
...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3
146	147	6.3	2.5	5.0	1.9
147	148	6.5	3.0	5.2	2.0
148	149	6.2	3.4	5.4	2.3
149	150	5.9	3.0	5.1	1.8

150 rows × 5 columns

In [20]:
```
dff.select_dtypes('object')  # display object values
```

Out[20]:

	Species
0	Iris-setosa
1	Iris-setosa
2	Iris-setosa
3	Iris-setosa
4	Iris-setosa
...	...
145	Iris-virginica
146	Iris-virginica
147	Iris-virginica
148	Iris-virginica
149	Iris-virginica

150 rows × 1 columns

In [22]:
```
group_species=dff.groupby('Species')
group_species
```

Out[22]:
```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x0000016AE5EB1C70>
```

In [23]:
```
group_species.first()
```

Out[23]:

Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	1	5.1	3.5	1.4	0.2
Iris-versicolor	51	7.0	3.2	4.7	1.4
Iris-virginica	101	6.3	3.3	6.0	2.5

In [24]:
```
group_species.count()
```

Out[24]:

Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	50	50	50	50	50
Iris-versicolor	50	50	50	50	50
Iris-virginica	50	50	50	50	50

In [25]:
```
group_species.mean()
```

Out[25]:

Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	25.5	5.006	3.418	1.464	0.244
Iris-versicolor	75.5	5.936	2.770	4.260	1.326
Iris-virginica	125.5	6.588	2.974	5.552	2.026

In [26]:
```
group_species.median()
```

Out[26]:

Species	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Iris-setosa	25.5	5.0	3.4	1.50	0.2
Iris-versicolor	75.5	5.9	2.8	4.35	1.3
Iris-virginica	125.5	6.5	3.0	5.55	2.0

In [29]:
```
grp=dff.groupby('Species')['SepalLengthCm'].agg(
    mean='mean',
    median='median',
    minimum='min',
    maximum='max',
    std_var='std',
    variance='var'
)
```

In [30]:
```
grp
```

Out[30]:

Species	mean	median	minimum	maximum	std_var	variance
Iris-setosa	5.006	5.0	4.3	5.8	0.352490	0.124249
Iris-versicolor	5.936	5.9	4.9	7.0	0.516171	0.266433
Iris-virginica	6.588	6.5	4.9	7.9	0.635880	0.404343

In [32]:

```
grp=dff.groupby('Species')['SepalWidthCm'].agg(
    mean='mean',
    median='median',
    minimum='min',
    maximum='max',
    std_var='std',
    variance='var'
)
grp
```

Out[32]:

Species	mean	median	minimum	maximum	std_var	variance
Iris-setosa	3.418	3.4	2.3	4.4	0.381024	0.145180
Iris-versicolor	2.770	2.8	2.0	3.4	0.313798	0.098469
Iris-virginica	2.974	3.0	2.2	3.8	0.322497	0.104004

In [33]:

```
grp=dff.groupby('Species')['PetalLengthCm'].agg(
    mean='mean',
    median='median',
    minimum='min',
    maximum='max',
    std_var='std',
    variance='var'
)
grp
```

Out[33]:

Species	mean	median	minimum	maximum	std_var	variance
Iris-setosa	1.464	1.50	1.0	1.9	0.173511	0.030106
Iris-versicolor	4.260	4.35	3.0	5.1	0.469911	0.220816
Iris-virginica	5.552	5.55	4.5	6.9	0.551895	0.304588

In [34]:

```
grp=dff.groupby('Species')['PetalWidthCm'].agg(
    mean='mean',
    median='median',
    minimum='min',
    maximum='max',
    std_var='std',
    variance='var'
)
grp
```

Out[34]:

Species	mean	median	minimum	maximum	std_var	variance
Iris-setosa	0.244	0.2	0.1	0.6	0.107210	0.011494
Iris-versicolor	1.326	1.3	1.0	1.8	0.197753	0.039106
Iris-virginica	2.026	2.0	1.4	2.5	0.274650	0.075433

In [ ]:

```
```